

HypatiaX Submission — Remediation Plan

Execution plan to address findings from the independent audit (cross-check of jmlr_paper_main.pdf, Supp. A, Supp. B, and the audit report)

Priority 1 — Blocking Issues (must fix before resubmission)

1. Figures 3 & 4 are broken and contradict Table 6

Root cause: the figure-generation pipeline outputs raw R^2 values (including NaNs and overflow from near-zero denominators) instead of handling them before rendering. Clipping is being applied after corruption already occurred, which is why cells labeled "clipped to $[-1, 1]$ " still show astronomical values.

Fix steps:

- 1 Pull the Table 6 JSON (the authoritative numeric source) rather than regenerating from the same broken pipeline that produced the figures.
- 2 Add explicit NaN/inf handling **before** the clipping step in the plotting code.
- 3 Regenerate Figures 3 & 4 directly from Table 6 values.
- 4 Spot-check the previously bad cells (Michaelis-Menten, Logistic Growth, Value at Risk, Gravitational Force, DeFi-AMM/Hard) against Table 6 one by one.
- 5 Run a full diff of every heatmap cell vs. its Table 6 counterpart — not just the ones the audit sampled.

2. Section 10.7 "reduction" paragraph is numerically backwards

The text claims a "reduction" from 0.678 to 0.789, but $0.789 > 0.678$ (an increase), and 0.678 does not correspond to any figure derivable from the paper's own numbers ($9/30 = 0.300$, or $142/180 = 0.789$).

Fix steps:

- 1 Identify what the 0.678 figure was actually supposed to represent — check whether it's a typo, a stray number from a different comparison, or an unresolved placeholder.
- 2 Confirm the correct primary result ($9/30 = 0.300$, matching abstract/Table 7) and the correct PCA-split corrected result ($142/180 = 0.789$).
- 3 Rewrite the paragraph with verified numbers, and align the direction-of-change language with the actual arithmetic.
- 4 Have someone outside the writing team read the rewritten paragraph in isolation to confirm it's self-consistent.

Draft replacement (author input still required — see notes):

"The increase from the random-split baseline ([INSERT VERIFIED VALUE] — see [source/table]) to the PCA-directed split result ($142/180 = 0.789$) reflects [INSERT: why a harder split would still show a numerically higher score — e.g., a different metric, a different eval subset, or a correction to an earlier miscalculation]. This is distinct from the primary result reported in the abstract and Table 7 ($9/30 = 0.300$), which uses [INSERT: clarify relationship between the two evaluations if they are not directly comparable]."

Open items the authors must resolve (not determinable from the audit alone): (1) the true, verified random-split baseline number, since 0.678 appears nowhere else in the paper; (2) whether "PCA split is harder" is still the correct explanation once real numbers are in — if the corrected trend goes baseline → higher score, that reasoning needs replacing, not relabeling.

Priority 2 — Real but Smaller Fixes

3. Supp. B Table 4 disagrees with abstract, Table 9, and Section 14 (HLLMNN: 90% vs. 100%)

- 1 Determine which threshold Table 4 actually used to compute 30/30 — its own footnote suggests a looser cutoff ($R^2 \geq 0.995$) rather than the stated $R^2 \geq 0.9999$.
- 2 Either correct Table 4's number to 27/30 (if the strict threshold applies), or relabel the column header to reflect the looser threshold actually used.
- 3 Confirm the fix doesn't require changes elsewhere — abstract, Table 9, and Section 14 already agree at 90% / 27/30, with the three named failures (Lorentz force, Photon energy, Zeeman energy).

4. Benchmark denominator drift in Supp. A (66 vs. 74)

- 1 Add a one-line reminder to the top of every Supp. A table: "n = 66, pre-v3.0 historical — see main paper Table 2 for the v3.0/n=74 confirmed figures."
- 2 Add a short explanatory sentence in Supp. A's introduction clarifying why the older denominator is retained, so readers don't need to reach Remark 1 first.

Priority 3 — Process Fix (Audit Tooling)

5. Audit report (HypatiaX_Paper_Audit_Report_Jul09.html) is non-functional

All six notebooks (NB-01–NB-06) failed to run, yet build_report.py generated a stale "all pass" summary ("0 open issues", 14 items marked RESOLVED) instead of surfacing the failure. This report should not be cited as evidence the paper is clean.

- 1 File this as a bug: build_report.py should hard-fail — not emit a green summary — when notebooks are missing.
- 2 Re-run NB-01 through NB-06 for real once the pipeline is fixed; treat the current HTML report as void until then.
- 3 Do not cite the current report as verification in any response to reviewers.

Suggested Execution Order

#	Task	Owner	Status
1	Regenerate Figures 3 & 4 from Table 6 JSON		■
2	Rewrite Section 10.7 reduction paragraph		■
3	Correct Supp. B Table 4 (HLLMNN threshold/number)		■
4	Add denominator reminders to Supp. A tables		■
5	Fix audit tooling (build_report.py) and re-run NB-01-06		■
6	Final pass: re-verify all previously-consistent figures still hold		■

Step 6 matters: edits to Figures 3/4 or Section 10.7 could accidentally disturb nearby text that currently checks out correctly — e.g., the 1.73× speedup figure, the 89.2% / 62.2% comparison, or the difficulty-tier breakdown (Easy 24 / Medium 29 / Hard 21).

Reference — Items That Already Check Out (do not need fixing)

- 1.73× LLM-routed speedup (n=68/74) — consistent across abstract, Contributions, §8.3, §10.4, Table 4, Table 11, Conclusion, and Supp. A.
- 89.2% near-perfect success rate, +27.0 pp over LLM, +83.8 pp over neural net — arithmetic checks out and figure repeats identically across abstract, Table 2, Table 3, §10.2, Supp. A.
- Difficulty-tier breakdown (Easy 24 / Medium 29 / Hard 21 = 74) and associated gains — match Table 3 exactly.
- Portfolio Variance seed-sweep arithmetic (Table 5) — means and win/recovery counts check out.
- 6 catastrophic LLM failures / 0 for HypatiaX; Michaelis-Menten far- $R^2 \approx -635$ vs. Table 6's -634.5989 — consistent.
- Table 1 vs. Supp. B Appendix G — Core-15 1231% mean neural extrapolation error, $U=0$, $p=1.11 \times 10^{-11}$ — matches exactly.
- Self-corrections (runtime claim, Portfolio Variance seed=42 attribution, Newton's-gravity threshold bug) — stated once, referenced consistently, no version drift.

Note: this plan is based on the independent audit findings only. Numeric corrections marked [INSERT] require the original authors/data to resolve — they could not be derived from the audit documents alone.